



PROJECT REPORT + KT

COVER PAGE

PUBG ANALYTICS DASHBOARD

Match Performance and Weapon Performance Analysis using Microsoft Power BI

Project Report and Knowledge Transfer

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Subject: **mini Project**

Academic Year: **2025–2026**

ABSTRACT

The **PUBG Analytics Dashboard** is a data visualization project developed using **Microsoft Power BI**. The project analyzes two major areas of PUBG data: **match performance** and **weapon performance**.

The Match Analytics Dashboard provides insights into the number of players, average kills, total matches, average survival time, average damage, match distribution, travel distance, damage dealt, and top-performing players.

The Weapon Analytics Dashboard analyzes the performance of different weapon types based on damage, range, rate of fire, DPS, and the number of shots required to eliminate a player through chest and head hits.

The main purpose of this project is to convert raw PUBG data into meaningful and interactive visualizations that make data analysis easier and more understandable.

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1. INTRODUCTION

Data visualization is the graphical representation of data using charts, graphs, tables, and dashboards. It helps users understand large amounts of information easily and identify patterns and trends.

PUBG generates a large amount of data related to players, matches, weapons, damage, kills, survival time, and other gameplay statistics.

In this project, Microsoft Power BI is used to analyze PUBG match and weapon data. The data is represented through interactive dashboards containing KPI cards, bar charts, donut charts, and other visualizations.

The project is divided into two main dashboards:

1. PUBG Match Analytics

This dashboard focuses on player and match performance.

2. PUBG Weapon Analytics

This dashboard focuses on weapon performance and comparison.

2. PROBLEM STATEMENT

Raw gaming data contains a large number of records and values that are difficult to analyze manually.

The problem is to transform PUBG match and weapon data into a simple and interactive dashboard that can provide useful insights about:

- Player performance
- Match performance
- Damage dealt
- Survival time
- Weapon damage
- Weapon range
- Rate of fire
- Damage per second

- Chest and head shots required to eliminate a player

Therefore, a Power BI dashboard is developed to visualize and analyze this information effectively.

3. OBJECTIVES

The main objectives of this project are:

- To analyze PUBG player and match performance.
 - To analyze the distribution of different match types.
 - To compare the damage dealt in different match types.
 - To analyze average survival time.
 - To identify top players based on kills.
 - To compare different weapon types based on damage.
 - To analyze the range of different weapons.
 - To compare weapon types based on DPS.
 - To analyze the rate of fire of weapons.
 - To compare the number of chest and head shots required to eliminate a player.
 - To present the analysis using an interactive Power BI dashboard.
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4. SCOPE OF THE PROJECT

The scope of this project includes:

- Analysis of PUBG match data.
- Analysis of PUBG weapon data.
- Creation of interactive visualizations.
- Comparison of different match types.
- Comparison of different weapon categories.
- Identification of performance trends.
- Presentation of results using a professional dashboard.

The dashboard can be useful for understanding gameplay statistics and comparing the performance of different match types and weapons.

5. DATASET DESCRIPTION

5.1 Match Dataset

The Match Dataset contains information related to PUBG matches and player performance.

Important fields include:

- Player ID
- Match ID
- Match Type
- Kills
- Damage Dealt
- Survival Time
- Walk Distance
- Ride Distance

This dataset is used to analyze player and match performance.

5.2 Weapon Dataset

The Weapon Dataset contains information about different PUBG weapons.

Important fields include:

- Weapon Name
- Weapon Type
- Damage
- Rate of Fire
- Range
- Damage Per Second (DPS)
- Shots to Kill – Chest
- Shots to Kill – Head

This dataset is used to compare weapon performance.

6. TOOLS AND TECHNOLOGIES

Microsoft Power BI

Microsoft Power BI is a business intelligence and data visualization tool used to create interactive reports and dashboards.

It was used in this project for:

- Importing data
- Creating visualizations
- Creating KPI cards
- Creating charts
- Applying formatting
- Creating interactive dashboards

Dataset

The PUBG match and weapon datasets were used as the source of data for analysis.

Data Visualization

Different visualizations were used to represent the data, including:

- KPI Cards
- Bar Charts
- Donut Charts
- Horizontal Bar Charts

7. METHODOLOGY

The project was completed using the following steps:

Step 1: Data Collection

The PUBG match and weapon datasets were collected for analysis.

Step 2: Data Import

The datasets were imported into Microsoft Power BI.

Step 3: Data Understanding

The fields and columns of the datasets were studied to understand the available information.

Step 4: Data Visualization

Suitable charts and KPI cards were selected based on the type of data.

Step 5: Dashboard Design

The dashboard was designed using a dark theme with orange highlights.

Step 6: Analysis

The visualizations were analyzed to identify trends and patterns.

Step 7: Presentation

The final dashboards were created to present the insights in an understandable format.

8. DASHBOARD PAGE 1: PUBG MATCH ANALYTICS

Dashboard Title: PUBG Match Analytics Dashboard

Description

The PUBG Match Analytics Dashboard provides an overview of player and match performance.

KPI Cards

Total Players

Displays the total number of players present in the dataset.

Average Kills

Shows the average number of kills achieved by players.

Total Matches

Displays the total number of matches available in the dataset

Average Survival Time

Shows the average survival time of players.

Average Damage

Displays the average damage dealt by players.

Visualizations

Travel Distance by Match Type

This chart compares the average walking distance and riding distance for different match types.

It helps understand how player movement varies between Solo, Duo, and Squad matches.

Damage Dealt by Match Type

This chart compares the amount of damage dealt in different match types.

It helps identify which match type contributes to the highest total damage.

Match Distribution

The donut chart shows the distribution of matches across different match types.

It helps understand the proportion of different match categories in the dataset.

Average Survival Time

This chart compares the average survival time for different match types.

It helps identify differences in player survival across various match modes.

Top 10 Players by Kills

This horizontal bar chart displays the top players based on their number of kills.

It helps identify the highest-performing players in the dataset.

9. DASHBOARD PAGE 2: PUBG WEAPON ANALYTICS

Dashboard Title: PUBG Weapon Analytics Dashboard

Description

The PUBG Weapon Analytics Dashboard analyzes the performance and characteristics of different weapons.

KPI CARDS

Total Weapon

Displays the total number of weapons in the dataset.

Average Damage

Shows the average damage value of the weapons.

Average Rate of Fire

Displays the average firing rate of weapons.

Average Range

Shows the average range of the weapons.

Average DPS

Displays the average Damage Per Second of the weapons.

VISUALIZATIONS

Damage by Weapon Type

This chart compares the damage values of different weapon types.

It helps identify weapon categories that provide higher damage.

Weapon Type Distribution

The donut chart displays the distribution of weapons across different weapon types.

It helps understand how many weapons belong to each category.

Range by Weapon Type

This chart compares the range of different weapon categories.

It helps identify weapon types that are more suitable for long-distance combat.

DPS by Weapon Type

DPS stands for **Damage Per Second**.

This chart compares the damage output per second of different weapon types.

A higher DPS indicates that a weapon can potentially deal more damage over a period of time.

Chest vs Headshots to Kill

This chart compares the number of shots required to eliminate a player using chest hits and headshots.

It demonstrates the difference in weapon effectiveness based on hit location.

Generally, headshots are more effective than chest shots because they cause greater damage.

10. KEY FINDINGS

The main findings from the project are:

1. The Match Analytics Dashboard provides a clear overview of player and match performance.
2. Different match types show variations in damage dealt, survival time, and travel distance.
3. The Match Distribution chart shows the proportion of different match types.
4. The Top 10 Players chart helps identify high-performing players based on kills.
5. The Weapon Analytics Dashboard shows that weapon types differ significantly in damage and range.
6. Different weapon categories have different DPS values.
7. Some weapon types are more suitable for long-range combat because of their higher range.
8. The chest versus headshot comparison shows that the number of shots required can vary based on hit location.
9. KPI cards provide a quick summary of important performance metrics.

10. Power BI makes the analysis more interactive and easier to understand.

11. ADVANTAGES OF THE PROJECT

- Easy to understand.
 - Interactive dashboard.
 - Provides visual analysis.
 - Helps compare different match types.
 - Helps compare different weapon categories.
 - Provides quick access to important KPIs.
 - Converts raw data into meaningful information.
 - Makes data analysis faster and more effective.
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12. LIMITATIONS

The project has some limitations:

- The analysis depends on the quality of the dataset.
 - The dataset may not represent every possible PUBG gameplay situation.
 - Player skill and strategy are not fully considered.
 - Weapon performance may vary depending on the game version and conditions.
 - The dashboard focuses mainly on the available fields in the datasets.
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13. FUTURE SCOPE

The project can be improved in the future by:

- Adding more recent PUBG datasets.
- Adding filters for specific weapons.
- Adding player-level performance analysis.
- Adding map-wise analysis.

- Adding time-based performance trends.
 - Adding predictive analytics.
 - Adding machine learning models to predict player performance.
 - Adding more interactive slicers and drill-through pages.
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14. CONCLUSION

The PUBG Analytics Dashboard successfully demonstrates the use of Microsoft Power BI for data analysis and visualization.

The project analyzes PUBG match performance and weapon performance using interactive dashboards.

The Match Analytics Dashboard provides insights into player performance, match distribution, damage, survival time, travel distance, and kills.

The Weapon Analytics Dashboard provides insights into weapon damage, range, rate of fire, DPS, weapon distribution, and chest versus head shots required to eliminate a player.

Overall, the project demonstrates how raw data can be converted into meaningful visual insights using data visualization techniques.

KNOWLEDGE TRANSFER / KT

Q1. What is Power BI?

Answer:

Power BI is a business intelligence and data visualization tool developed by Microsoft. It allows users to connect to data, analyze it, and create interactive reports and dashboards.

Q2. What is a dashboard?

Answer:

A dashboard is a visual representation of important data and information. It contains charts, graphs, and KPI cards that help users understand data quickly.

Q3. What is a dataset?

Answer:

A dataset is a collection of related data organized in the form of rows and columns. In this project, PUBG match and weapon data were used.

Q4. Why did you choose PUBG data for this project?

Answer:

PUBG data contains different performance metrics such as kills, damage, survival time, weapon damage, range, and DPS. This makes it suitable for data analysis and visualization.

Q5. What are the two main parts of your project?

Answer:

The project consists of:

1. PUBG Match Analytics
 2. PUBG Weapon Analytics
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Q6. What is a KPI card?

Answer:

A KPI card displays an important single value or performance indicator. Examples in this project include Total Players, Average Damage, Total Weapons, and Average DPS.

Q7. Why did you use KPI cards?

Answer:

KPI cards provide a quick summary of important values. Users can understand the main performance metrics without analyzing the entire dashboard.

Q8. Why did you use a bar chart?

Answer:

Bar charts are useful for comparing values across different categories. In this project, they are used to compare damage, range, DPS, and other metrics between weapon types.

Q9. Why did you use a donut chart?

Answer:

A donut chart is useful for showing the proportion or distribution of categories. It is used to display match distribution and weapon type distribution.

Q10. What is a slicer?

Answer:

A slicer is an interactive filter in Power BI. It allows users to select specific values and filter the dashboard accordingly.

Q11. What is the purpose of the Match Analytics Dashboard?

Answer:

The purpose is to analyze player and match performance using metrics such as kills, damage, survival time, travel distance, and match type.

Q12. What is the purpose of the Weapon Analytics Dashboard?

Answer:

The purpose is to compare different weapons and weapon types based on damage, range, rate of fire, DPS, and shots required to kill.

Q13. What is DPS?

Answer:

DPS stands for **Damage Per Second**. It represents the amount of damage a weapon can deal per second.

Q14. Why did you compare chest shots and headshots?

Answer:

Chest shots and headshots cause different levels of damage. Comparing them helps understand how many shots are required to eliminate a player based on the hit location.

Q15. What is the difference between Sum and Average?

Answer:

Sum is the total of all values, while Average is calculated by dividing the total value by the number of values.

Q16. What is data visualization?

Answer:

Data visualization is the graphical representation of data using charts, graphs, and other visual elements.

Q17. What is the difference between a report and a dashboard?

Answer:

A dashboard generally provides a high-level summary of important information, while a report can contain multiple pages and detailed analysis.

Q18. What is the purpose of the dark theme in your dashboard?

Answer:

The dark theme provides a gaming-style appearance suitable for PUBG. Orange highlights are used to improve contrast and maintain visual consistency.

Q19. What is the main advantage of Power BI in this project?

Answer:

The main advantage is that Power BI allows data to be transformed into interactive and visually appealing dashboards, making analysis easier.

Q20. What are the limitations of your project?**Answer:**

The analysis depends on the available dataset. It may not include all possible gameplay factors, and weapon performance can vary depending on different game conditions and updates.

Q21. What improvements can be made in the future?**Answer:**

Future improvements can include predictive analysis, machine learning models, map-wise analysis, player-level filtering, real-time data, and more advanced interactive features.
